

Sec. 3.4: More set operations

Definition. Let A and B be two sets.

- The **set difference** of A and B , written $A - B$, is the set of elements of A that is not in B .

Venn diagram:

- The **complement** of A , written $\bar{A}$, is the set of elements in U that are not in A , where U is an universal set made clear by the context.

Venn diagram:

- **Note:** $A - B = A \cap \bar{B}$, $\bar{\bar{A}} = A$

Example 1.

- If $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 5, 7, 9\}$, then $A - B = \{1, 2, 4\}$
 $B - A = \{7, 9\}$
- If $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{2, 4, 6, 8\}$, then $\bar{A} = \{1, 3, 5, 7\}$
- If $U = \mathbb{R}$, $A = (-\infty, 1) \cup (1, 3]$, then $\bar{A} = \{1\}$

Definition. Let A and B be two sets.

- The **symmetric difference** of A and B , written $A \oplus B$, is the set of elements that are in A or B , but not in both.

Venn diagram:

- Other ways to express $A \oplus B$:

Example 2.

- $\{a, b, c\} \oplus \{x, y, a\} = \{b, c, x, y\}$

De Morgan's Law for sets

• $\overline{A \cup B} =$

• $\overline{A \cap B} =$

Exercise 3. Let $U = \mathbb{Z}$ be the universal set. Define the following sets.

• $A = \{-3, -1, 0, 1, 2, 3, 4, 9, 10\}$ • $C = \{-3, 0, 1, 3, 5, 6\}$ • $E = \{x \in \mathbb{Z} : x \text{ is even.}\}$

• $B = \{0, -1, -2, -3\}$ • $D = \{x \in \mathbb{Z}^+ : x < 7\}$

(a) $(A - C) \cup B =$

(b) $A \oplus D =$

(c) $A \oplus B \oplus C =$

(d) $(A \cap B) - \overline{E} =$

(e) $(\overline{C \cap A}) \cap B =$

Summary for Sec. 3.3 and 3.4 - fill in the blanks. Answers will be posted on Canvas.

- Let A, B be two sets and U be the universal set. Complete the table about operations on sets:
*For "Example", use $A = \{1, 2\}, B = \{2, 3\}, U = \{1, 2, 3, 4\}$

Symbol	Name	Definition	Venn diagram	Example*
$A \cup B$		$x \in A \cup B \leftrightarrow$		
$A \cap B$		$x \in A \cap B \leftrightarrow$		
$A - B$		$x \in A - B \leftrightarrow$		
$\overline{A}$		$\forall x \in U, x \in A \leftrightarrow$		
$A \oplus B$		$x \in A \oplus B \leftrightarrow$		

• $(A_1 \cup A_2) \cup A_3 =$, $(A_1 \cap A_2) \cap A_3 =$

• $A \cap (B \cup C) =$, $A \cup (B \cap C) =$

• $\overline{A \cup B} =$, $\overline{A \cap B} =$